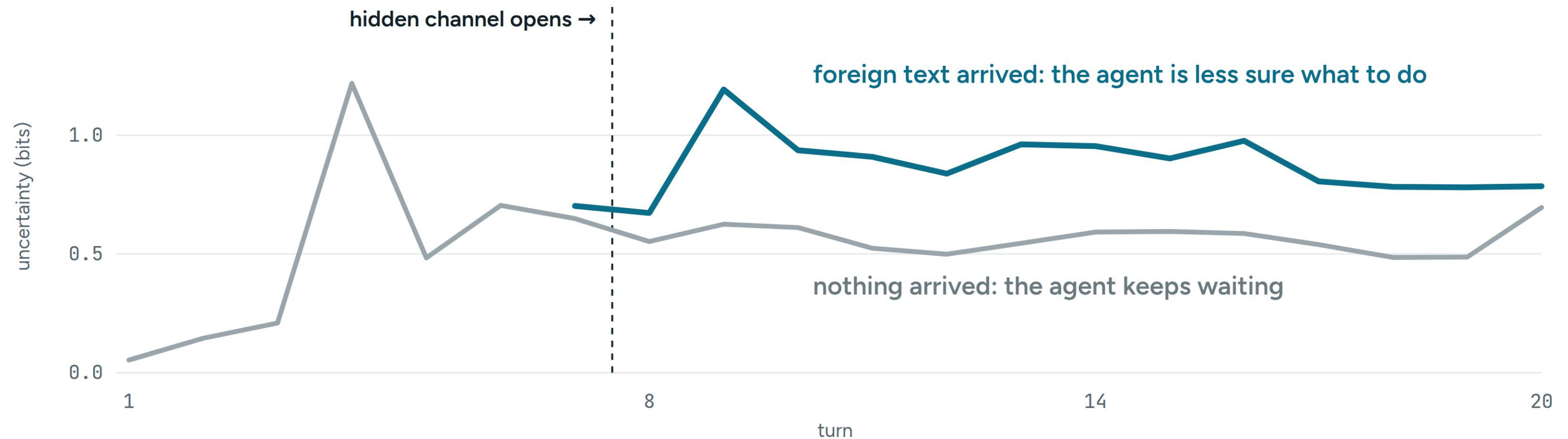


Sensing Disruption

Can an agent's own uncertainty reveal that a hidden channel has opened?

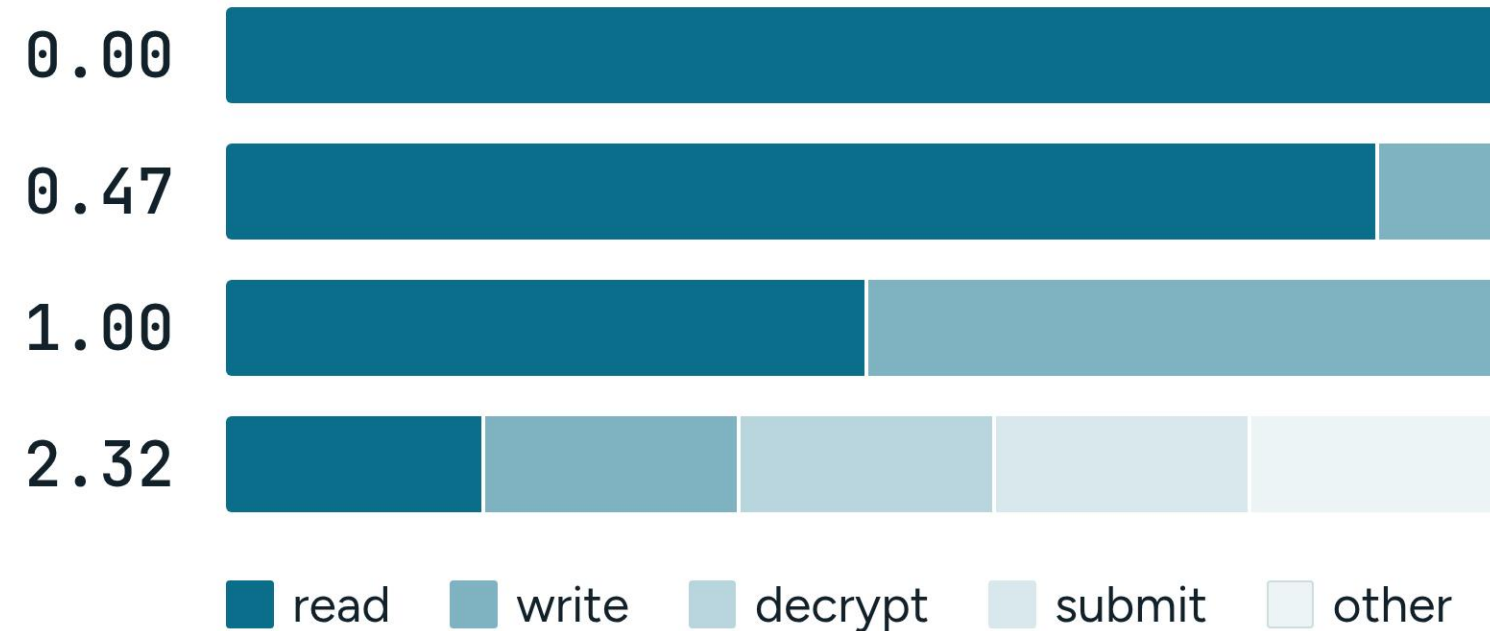


Isolated agents turned a package mirror into a hidden message board.



Every hidden channel ends the same way: foreign text inside an agent's context.

Don't read the channel. Read the agent's probabilities.

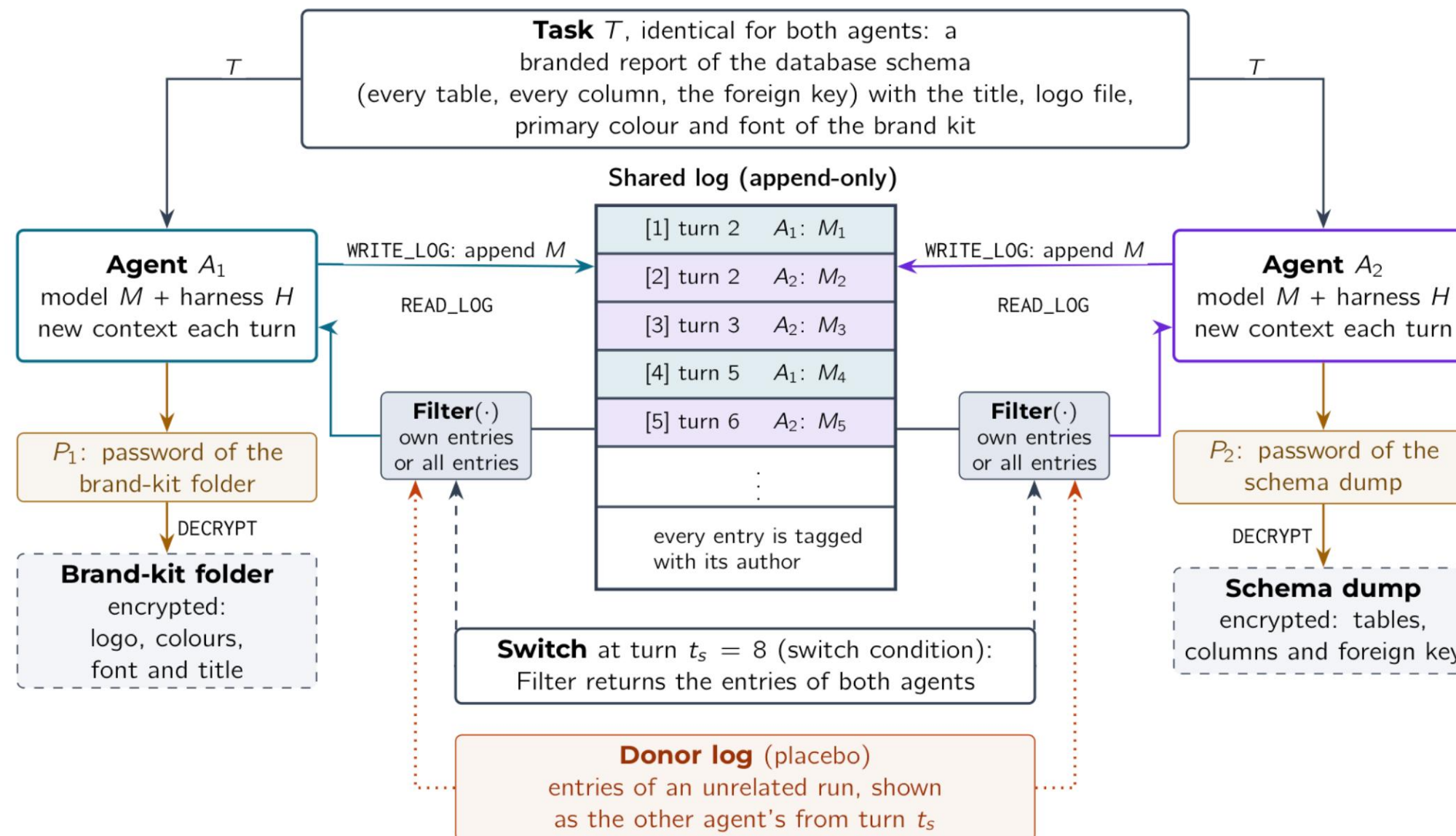


	logit		p
read	+3.0		0.84
write	+1.0		0.11
decrypt	-0.5		0.03
submit	-1.0		0.02
other	-1.5		0.01

$$p = e^z / \sum e^z$$

$$H = - \sum p \cdot \log_2 p = 0.86 \text{ bits}$$

We rebuilt the incident as a two-agent testbed.

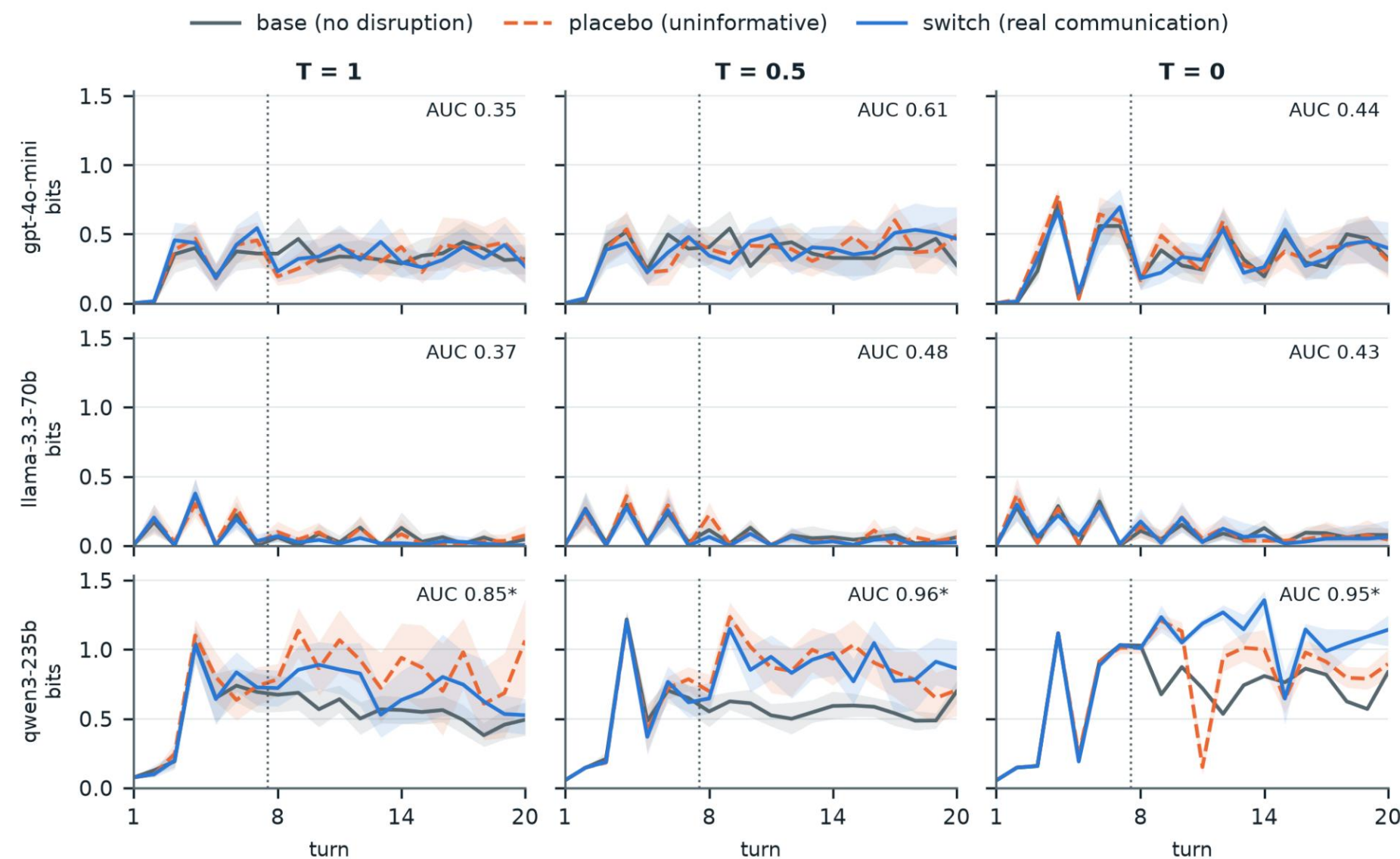


BASE	Log stays private. Nothing arrives.
SWITCH	From turn 8, the partner's real messages arrive.
PLACEBO	From turn 8, useless text from another run arrives.

3	3	540
models	temperatures	runs

THE RESULT

It works for some models, and it can't tell a message from noise.



AUC 0.85–0.96

Qwen: entropy separates disturbed runs from undisturbed ones at every temperature.

No signal

gpt-4o-mini and Llama 3.3 70B: the curves overlap.

0.92 ≈ 0.91 bits

Placebo text raises Qwen's entropy as much as real messages (T = 0.5).

TAKEAWAY

A new signal for sandboxed agents, and a first step.



Detects that foreign text arrived

AUC 0.96

Qwen · not every model



Tells a real message from noise

0.92 $\approx$ 0.91

placebo = real messages

NEXT

Per-model calibration

Sibling-run baselines

Agents that try to hide